

Advanced Probability

Notes based on Book: *Probability: Theory and Examples* by Rick Durrett

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Chapter 1: Measure Theory

1.1 Probability Spaces

一个**概率空间**是一个三元组 $(\Omega, \mathcal{F}, P)$ ，其中 Ω 是结果/Outcomes 的集合， $\mathcal{F}$ 是事件/Events 的集合，而 $P: \mathcal{F} \rightarrow [0, 1]$ 是一个给事件分配概率的函数。我们假设 $\mathcal{F}$ 是一个 **σ -代数**，即 Ω 的子集的一个非空集合，满足对可数交、可数并和补集运算的封闭性。仅给定 $(\Omega, \mathcal{F})$ ，我们称之为 **可测空间**，即我们可以通过其定义测度的空间，所谓测度是一个非负的、可数可加的函数，即一个函数 $\mu: \mathcal{F} \rightarrow \mathbb{R}$ ，满足

- 对所有 $A \in \mathcal{F}$ ， $\mu(A) \geq \mu(\emptyset) = 0$;
- 若 $A_i \in \mathcal{F}$ 是互不相交的集合序列，则 $\mu(\bigcup_i A_i) = \sum_i \mu(A_i)$ 。

如果对于全集 Ω ， $\mu(\Omega) = 1$ ，我们称 μ 是一个 **概率测度/Probability Measure**。在本书中，概率测度通常用 P 表示。

Theorem 1.1.1

令 μ 是 $(\Omega, \mathcal{F})$ 上的一个测度，则 μ 满足下面性质：

1. **单调性/Monotonicity**。如果 $A \subset B$ ，则 $\mu(A) \leq \mu(B)$;
2. **次可加性/Subadditivity**。如果 $A \subset \bigcup_{m=1}^{\infty} A_m$ ，则 $\mu(A) \leq \sum_{m=1}^{\infty} \mu(A_m)$;
3. **下连续性/Continuity from below**。如果 $A_i \uparrow A$

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5.3 Recurrence and Transience

5.4 Stationary Measures

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5.6 Periodicity, Tail σ -Field*

5.7 General State Space*

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